



Inflammation

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What is your expectations?

Objectives

- **Define inflammation**
- **Differentiate between acute and chronic inflammation**
- **Recognize the steps of extravasation of inflammatory cells**
- **Recognize the process of chemotaxis and phagocytosis**
- **Identify different examples of inflammation**

Definition

Local vascular and cellular response of
living tissue against an injurious agent

Terminology

- Inflammation of certain tissue → the tissue inflamed+ **itis**

e.g.

Gastritis

Colitis

Tonsillitis

Appendicitis

Aims of inflammation

- Fighting bacteria
- Localize infection
- Removal of damaged tissue

*** Causes of inflammation:**

(1) Living Irritants: Bacteria and their toxins, viruses, parasites and fungi.

(2) Non Living Irritants: include:

(a) *Physical irritants:* e.g. excess heat, excess cold and radiations.

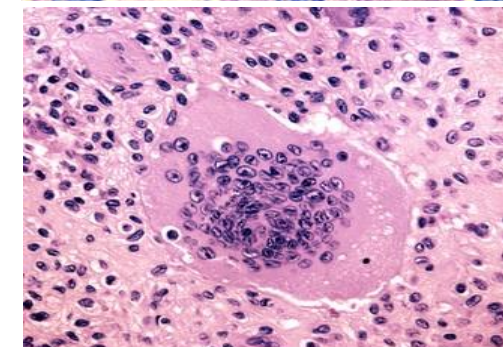
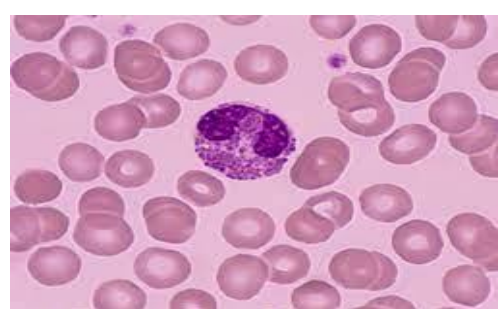
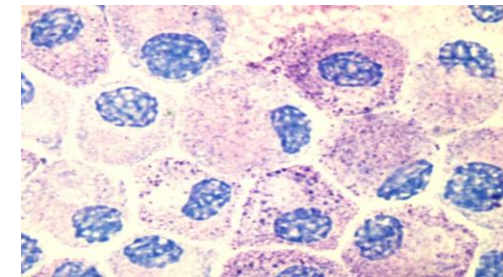
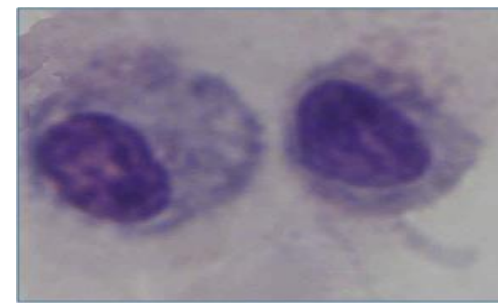
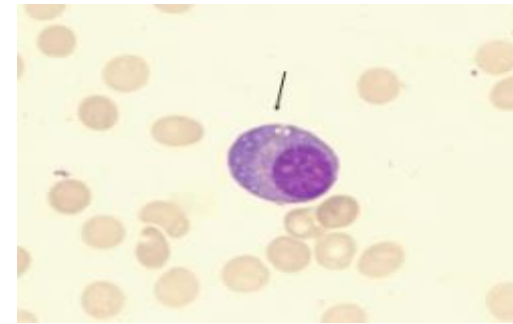
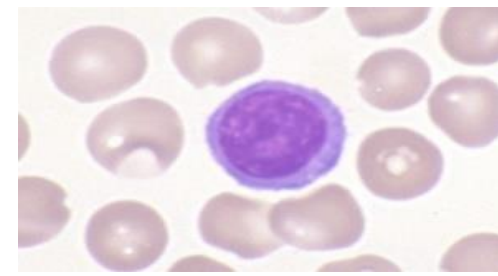
(b) *Chemical irritants:* e.g. concentrated acids, alkalis, poisons.

(c) *Mechanical irritants:* e.g. trauma, mechanical friction and foreign bodies.

(d) *Immunological:* e.g. allergic inflammation.

Inflammatory cells

- Neutrophils: phagocytosis, pus cells
- Lymphocytes: chronic
- Plasma: chronic
- Histiocytes: Chronic phagocytosis, monocytes
- Eosinophils: allergy, parasitic
- Mast cells: allergy (histamine, serotonin)
- Giant cells: phagocytosis, histiocytes



5 Cardinal Signs of Inflammation



Pain



Heat



Redness

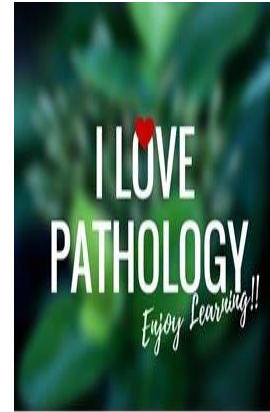


Swelling



**Loss of
Function**

- Fever
- Leukocytosis
- Malaise
- Nausea
- Anorexia
- Lymphoid hyperplasia
- CRP, ESR raised



INFLAMMATION

SYSTEMIC EFFECTS



@VijayPatho

Types of inflammation

1- Acute

2- Chronic

Acute & Chronic Inflammation

Features	Acute	Chronic
Onset	Fast: minutes to hours	Slow; days
Cellular infiltrate	Neutrophils mainly	Monocytes/macrophages & Lymphocytes
Tissue injury, fibrosis	Usually mild and self limited	Often severe and progressive
Local & Systemic signs	Prominent	Less prominent



ACUTE

INFLAMMATION



Main events of inflammation

- 1- Vascular response
- 2- Inflammatory fluid exudate
- 3- Inflammatory cellular exudate

1) Vascular Response

(1) Changes in Vascular Flow and Caliber

Vasodilatation (histamine)

Stasis (+ viscosity, swollen end., open capillaries)

(2) Increased Permeability and Formation of Fluid Exudate

Histamine, Kinins

Endothelial “gaps”

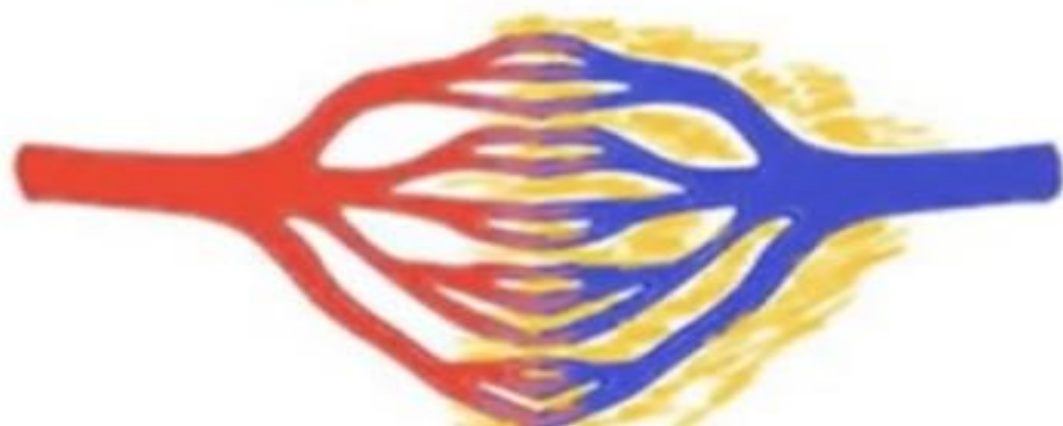
Direct endothelial Injury



1. Vasoconstriction (seconds)

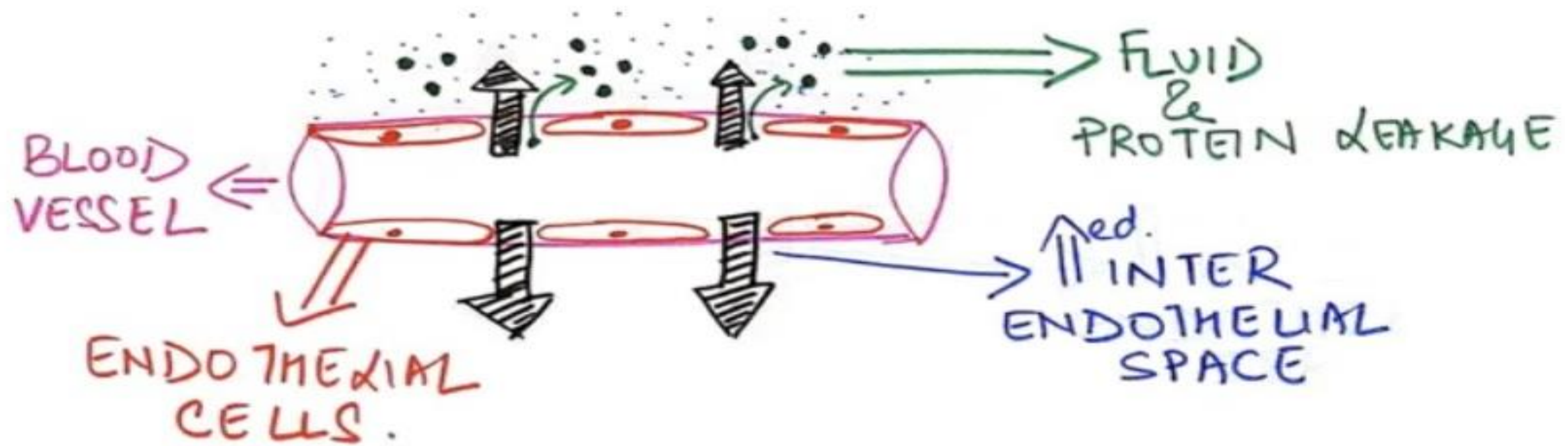


2. Vasodilation (minutes)



3. Permeability increase
- oedema formation

Vascular permeability



↑^{ed} VASCULAR PERMEABILITY

- ⇒ LOSS OF FLUID & ↑^{ed} VESSEL DIAMETER.
- ⇒ CONCENTRATION OF RBC'S
- ⇒ ↑^{ed} VISCOSITY OF BLOOD
- ⇒ SLOWER BLOOD FLOW
- ⇒ STASIS

2) Inflammatory fluid exudate

- **Formation:**

- 1-VD, +Permeability
- 2- + interstitial osmotic
- 3- -hydrostatic

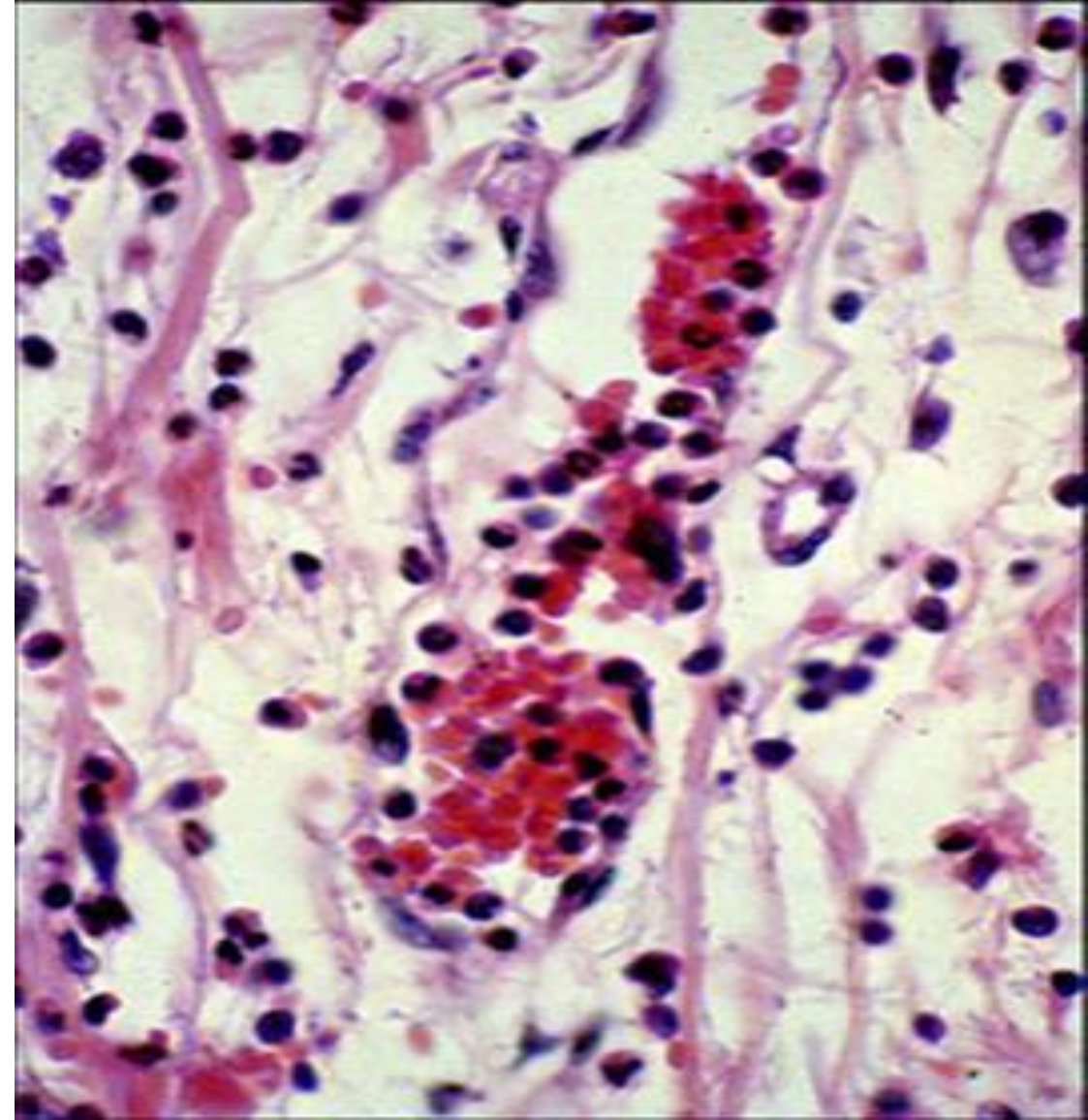
- **Function:**

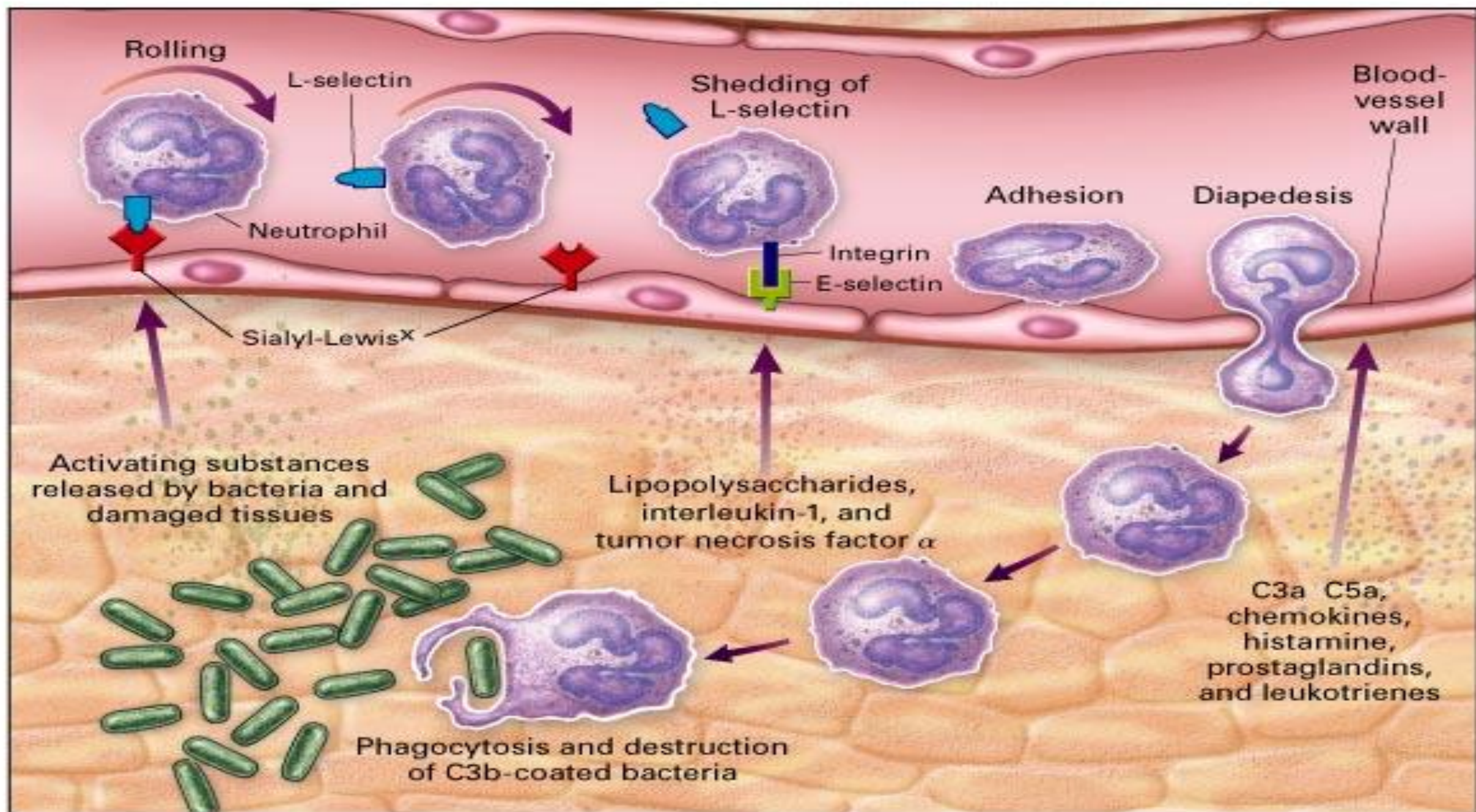
- Dilute toxin, chemical, poison
- Bring antibodies
- Supply nutrition for cells remove toxins
- Supply fibrinogen (localization, movement of inflammatory cells and fibroblasts)

3) Cellular exudate

Extravasation of neutrophils

- **MARGINATION** (PMN's go toward wall)
- **ROLLING** (tumbling and EAPING (selectin-glycoprotein))
- **ADHESION** (integrin-Ig)
- **TRANSMIGRATION**(DIAPYCNOSIS,(PCAM))





Chemotaxis

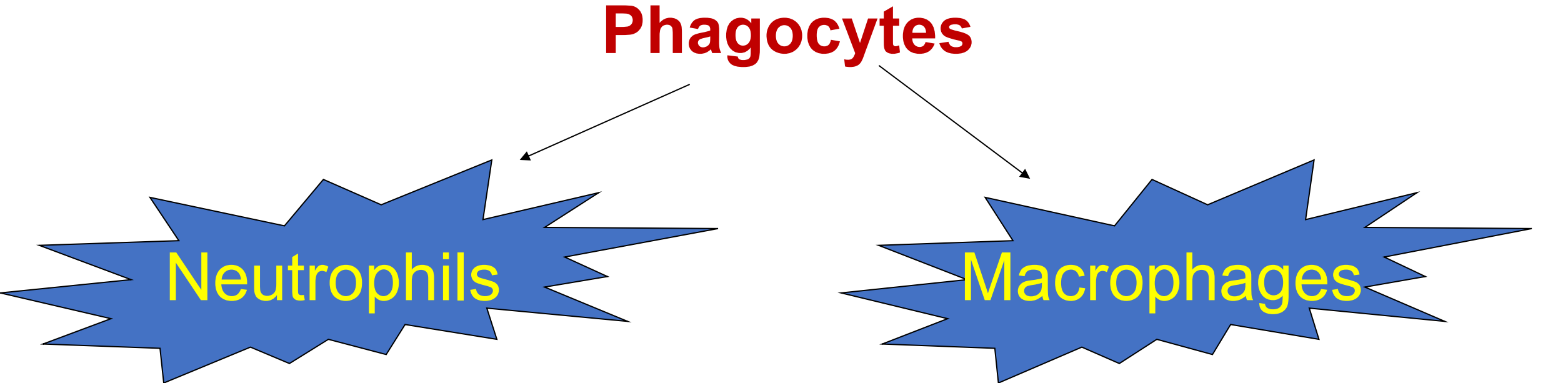
Definition: Is the directed movement of **neutrophils** and **macrophages** in the area of inflammation towards the irritant depending on

- 1-Exogenous: e.g. bacterial product
- 2-Endogenous: mediators as complement, arachidonic acid metabolites and chemokines. Those lead to polymerization of actin and facilitation of movement

Phagocytosis major benefit

Is the ingestion and destruction of bacteria, necrotic debris and foreign particles by the phagocytic inflammatory cells.

Phagocytes



The diagram illustrates the classification of phagocytes. At the top, the word "Phagocytes" is written in red. Two arrows point downwards from this word to two blue, starburst-shaped boxes. The left box contains the word "Neutrophils" in yellow, and the right box contains the word "Macrophages" in yellow.

Neutrophils

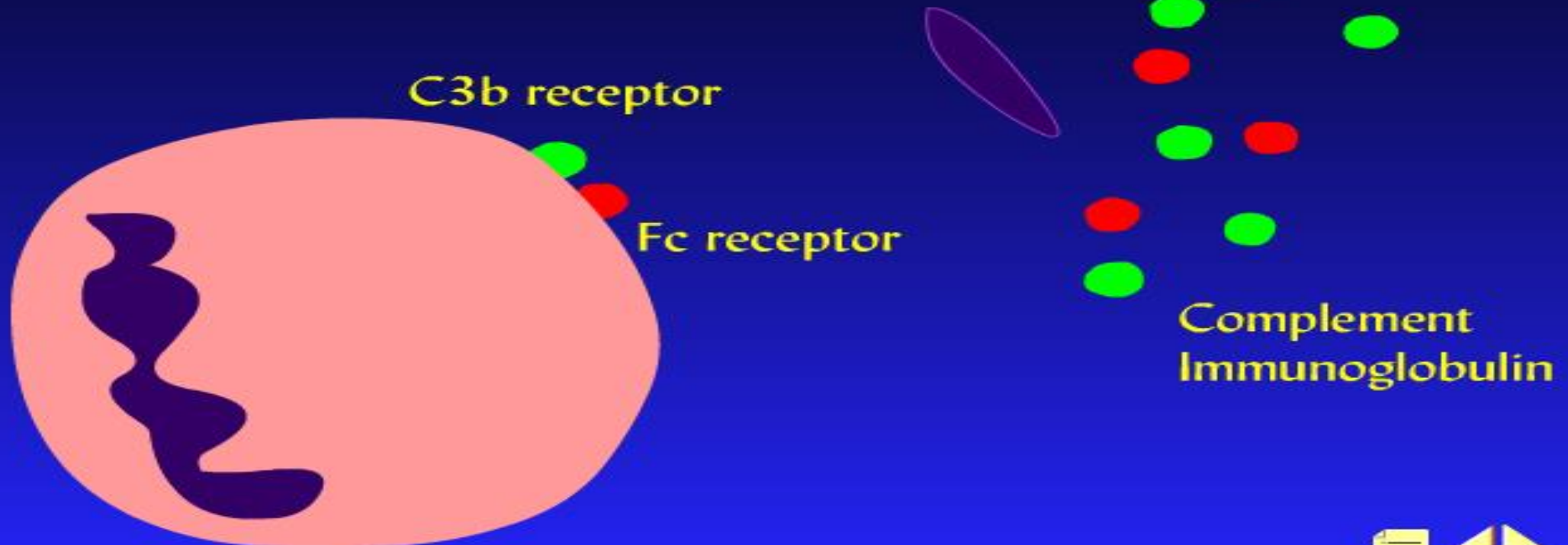
Macrophages

1- Recognition

Opsonin (Ig, complement)

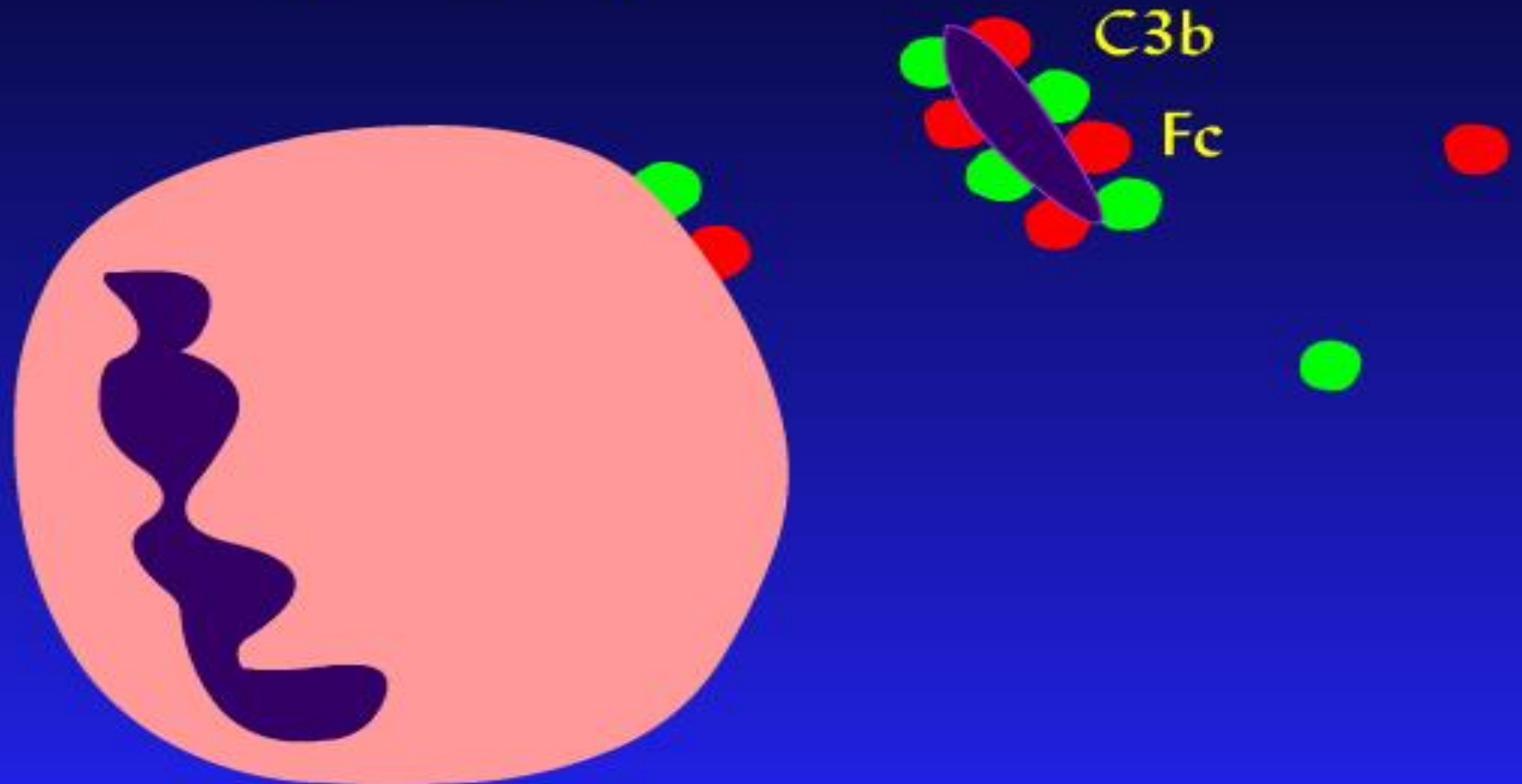
Cellular Responses

Opsonization and phagocytosis



Cellular Responses

Opsonization and phagocytosis



Cellular Responses

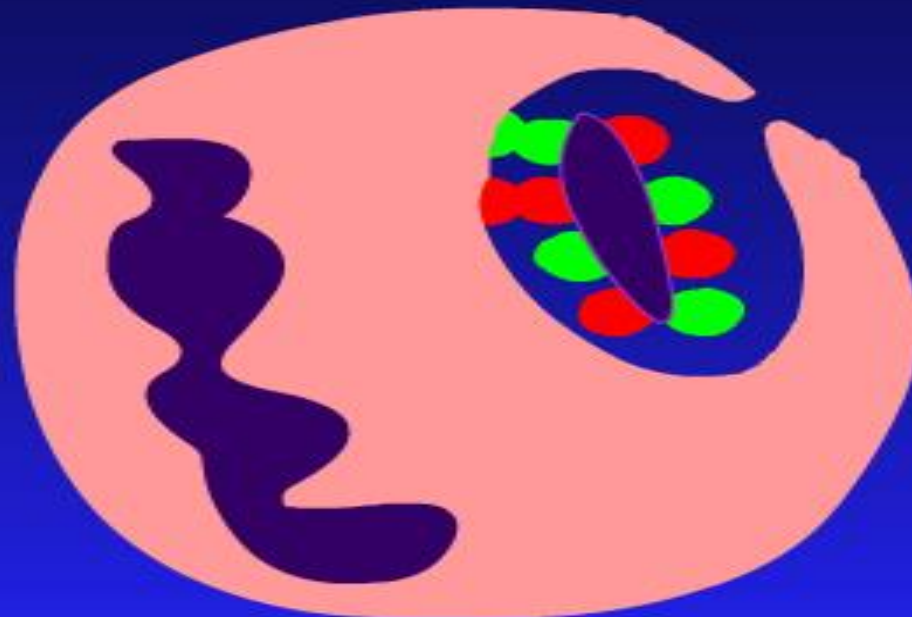
Opsonization and phagocytosis



2- Engulfment

Cellular Responses

Opsonization and phagocytosis



3- Killing and degradation (Oxidative and non-oxidative)

Cellular Responses

Oxidative burst, bacterial killing, and tissue injury



Outcomes of acute inflammation

```
graph LR; A[Outcomes of acute inflammation] --- B[Resolution]; A --- C[Regression & Healing]; A --- D[Progression & spread]; A --- E[Progression & Chronicity];
```

Resolution

Regression & Healing

Progression & spread

Progression & Chronicity

Summary Of Sequence of Events

- **Vasodilatation → increased vascular permeability → leakage of exudate, stasis → extravasation,** margination, rolling, adhesion, transmigration (diapedesis)
→ chemotaxis → PMN activation → phagocytosis → termination →
either 100% resolution, scar formation, or chronic inflammation.

Types of acute inflammation

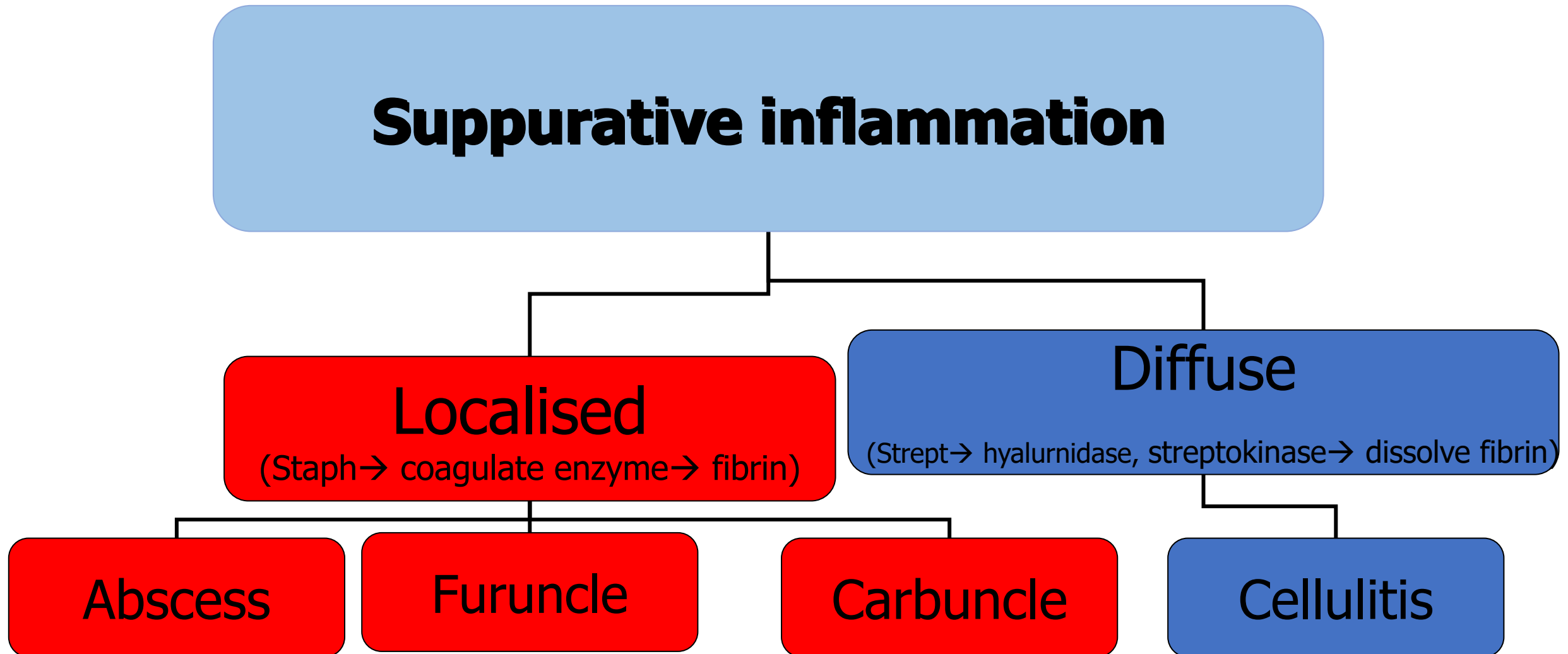
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graph TD; A[Acute inflammation] --> B[Suppurative]; A --> C[Non-Suppurative]
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Acute inflammation

Suppurative

Non-Suppurative

Types of Acute Suppurative inflammation



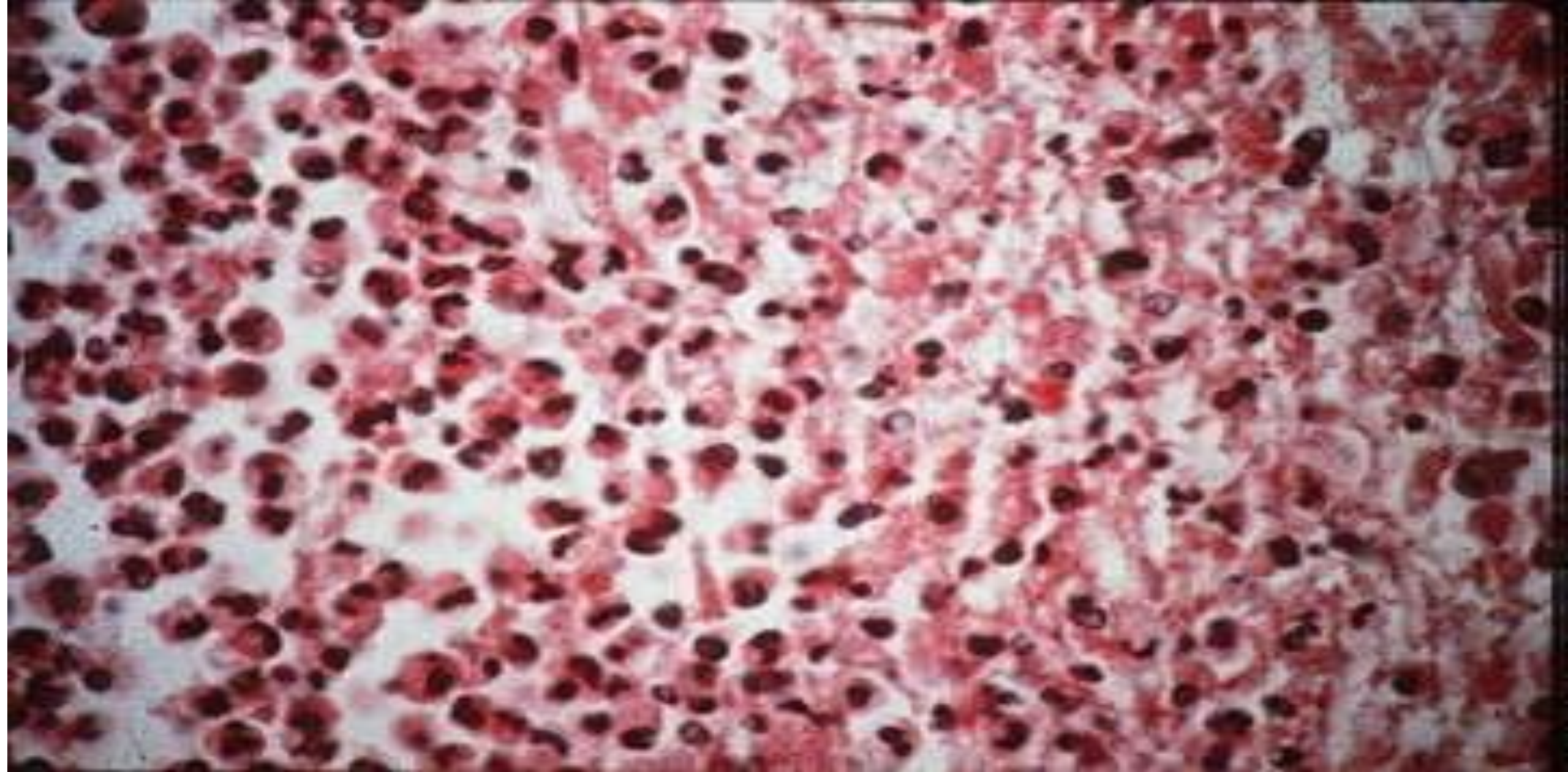
Abscess

- **Definition:** Localized suppurative acute inflammation characterized by formation of pus
- **Cause:** Staph → Excess fibrin
- **Site:** Skin (subcutaneous tissue), other organs
- **Clinical:** red, hot, swollen, Pus



PUS
=
PURULENT

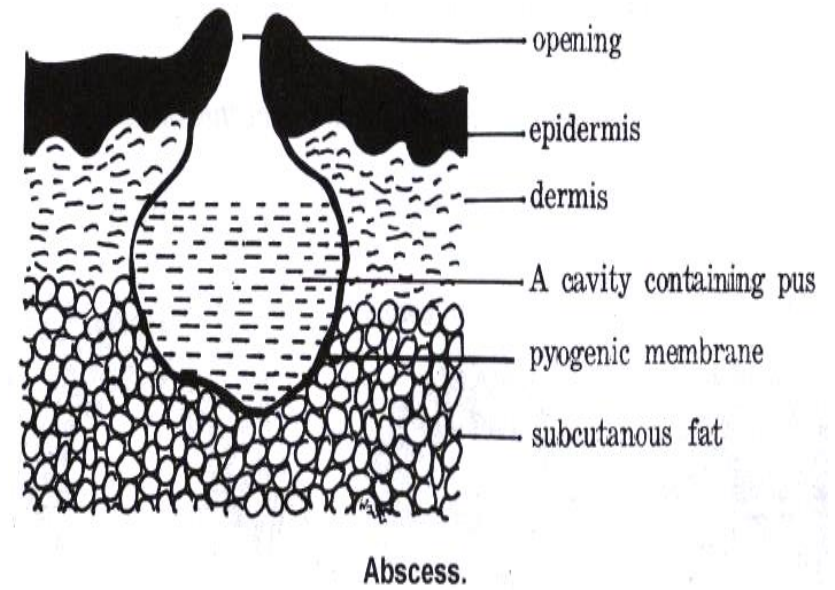
ABSCESS
=
POCKET
OF
PUS



PURULENT, FIBRINOPURULENT

Boil-Furuncle

Small abscess related to hair follicle or sebaceous gland

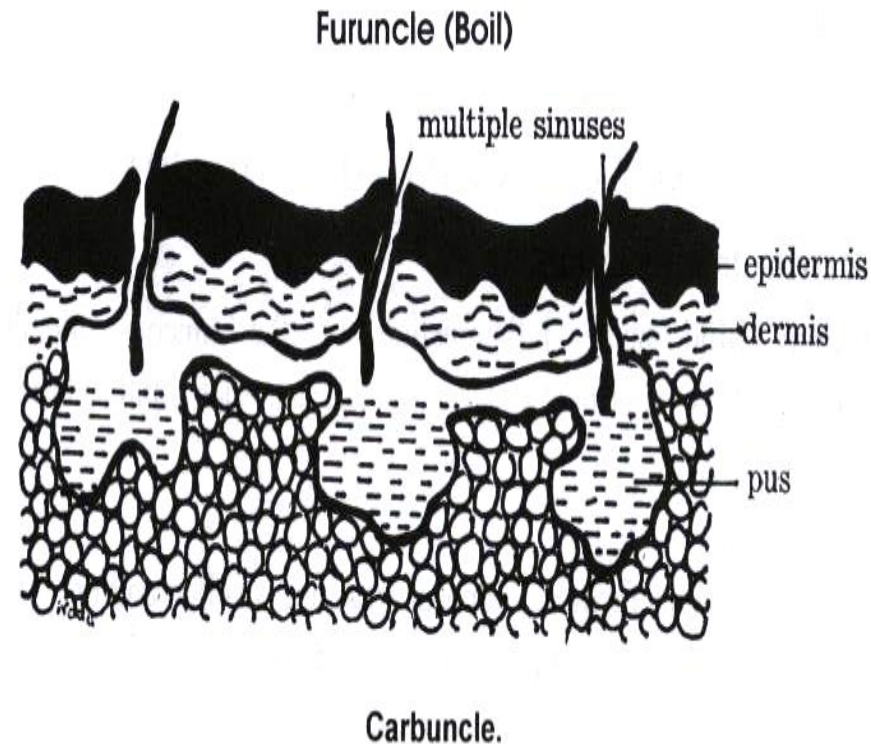


Carbuncle

Multiple communicating abscess

Risk: Diabetics (low immunity)

Site: back of neck (thick tough skin and subcutaneous tissues → dense fibrous septa → divide the fat into compartments)



Diffuse suppurative Inflammation

- Causative organism: streptococcus haemolyticus producing
- Enzyme 1-fibrinolysin, hyaluronidase and 2-spreading factor
- Examples:
 - Cellulitis
 - Appendicitis
 - Peritonitis

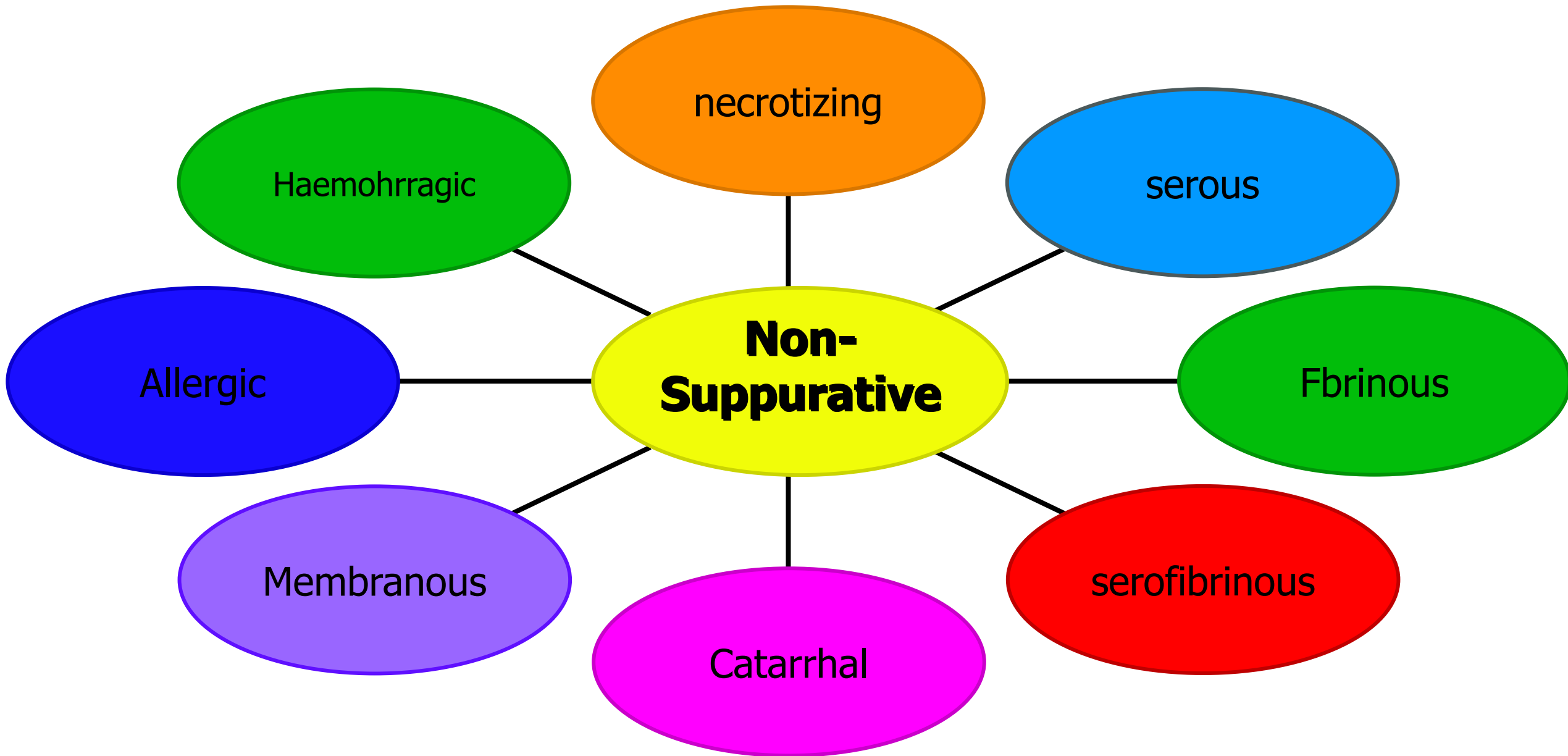


Purulent Exudate



	Abscess	Cellulitis
Def	Localized suppurative examination	Diffuse suppurative examination
Causative organism	Staph	Strept
Mechanism	Excess fibrin	fibrinolysin, hyalouronidase and spreading factor
Site	Any tissue	Loose tissue
Pus	Thick yellow	Thin sanguinous(bloody)
Spread	Less	More common

Types of Acute Non-Suppurative inflammation



Types of Acute Non-Suppurative inflammation

Serous inflammation: Excess watery fluid exudate → blisters in burn

- **Haemorrhagic inflammation:** excessive RBCs within the exudate → vascular damage. e.g chicken box
- **Allergic inflammation:** hypersensitivity → many eosinophils e.g. Urticaria

Types of Acute Non-Suppurative inflammation

Catarrhal inflammation: irritated mucous membrane → Rhinitis, excess mucous

Pseudoembranous inflammation: severe, mucous membranes, toxins of bacteria → necrosis, false membrane, Fibrin patches. E.g. Diphtheria

Thank
you!

A vibrant, cartoon-style graphic featuring the words "Thank you!" in a bold, bubbly, 3D font. The word "Thank" is rendered in a pink-to-orange gradient, while "you!" is in a blue-to-green gradient. The letters have thick black outlines and are surrounded by several yellow five-pointed stars and colorful, striped ribbons in shades of pink, orange, and blue. The overall design is playful and celebratory.